

Phishing Email Investigation

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Author Note

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Project Objectives

The objectives of this project were to:

- Analyze an email for phishing indicators
- Examine email headers for sender authenticity
- Validate SPF, DKIM, and DMARC authentication
- Inspect embedded hyperlinks
- Identify indicators of compromise
- Perform threat intelligence lookups
- Assess the overall security risk
- Produce a professional incident investigation report

Lab Environment

Item	Details
Operating System	macOS
Browser	Google Chrome
Email Sample	Simulated Microsoft 365 Phishing
Header Analysis	Google Admin Toolbox / MXToolBox
Threat Intelligence	VirusTotal, AbuseIPDB
Investigation Type	Phishing Email Analysis

Methodology

The investigation followed a structured email analysis process:

1. Examined the reported phishing email
2. Reviewed sender information
3. Analyzed email headers

4. Verified SPF authentication
5. Verified DKIM signatures
6. Verified DMARC policy results
7. Inspected embedded hyperlinks
8. Investigated sender IP reputation
9. Identified indicators of compromise
10. Performed a risk assessment
11. Documented findings and recommendations

Investigation Findings

5.1 Initial Email Review

A user reported receiving an email claiming that they would receive a gift from Vivinit, specifically a “Free Installation” + “\$50 Visa Gift Card”. The email instructed the recipient to click on a button to receive the offer.

The message itself contained a green checkmark emoji in the header and the image displaying the offer was blurry, the text was also very poorly selected.

Figure 1. Reported email submitted for investigation.



5.2 Email Header Analysis

The email headers were analyzed using the MXToolbox Email Header analyzer. Header analysis revealed several inconsistencies regarding the sender's identity.

The sender used the domain

nooreply@4695370485734.zbsbwglwokmebkigh.org.uk

Although the email states that they are from the “Vivint” company, the domain that was used is not associated with the company at all. Their actual email should be under

support@vivint.com or security@vivint.com

Figure 2. Email header analysis identifying sender information, routing path, and message metadata.

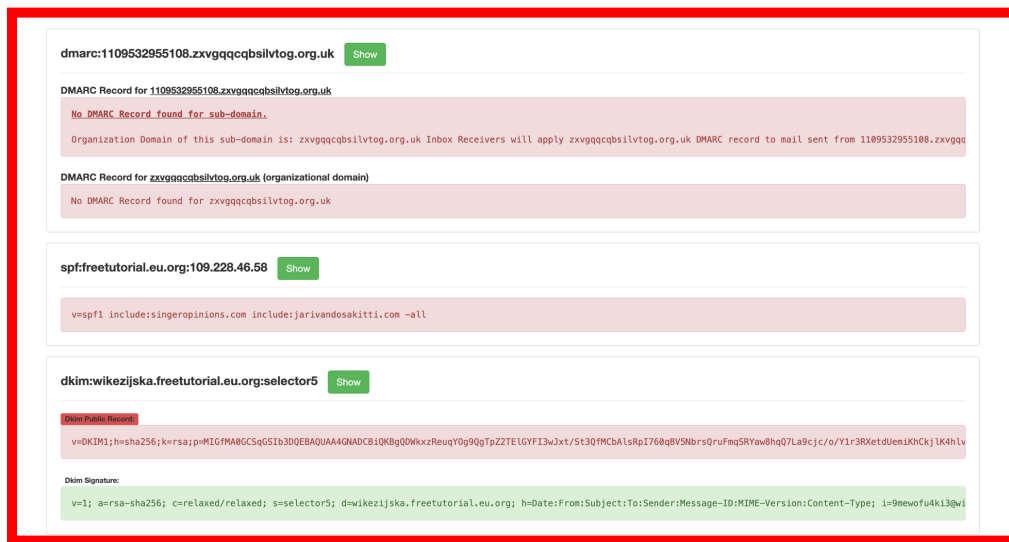
Header Name	Header Value
Delivered-To	
X-Received	by 2002:a5d:5f91:0:b0:472:d154:fac6 with SMTP id ffacd0b85a97d-47fc6666cdemr24173f8f.35.1785350324222; Wed, 29 Jul 2026 11:38:44 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=1785350324; cv=none; d=google.com; s=arc-20260327; b=c09ZSCVp68JLPY1PkDFJMJUN2cQ6aXMQMk4RgT9dJ8Qq52s+UkOW0ZQI96J91MRDg 5Jf2KwSvzjHxV4WVhTMF2Y3lxYVXZ0B V6kKcAzJ16HZJdnFJUVpUkVvYUeFgrt If6ZIF7P+7Gb3QZeMUUV5GE6zDkR2H/bnL+njDqHkYt2GElc1000/JSySC7evy/ Souy2SVHbtdHm14N7WhSwdPZUwqYF21wqIDGJDCH3vkzMTlySQXg260CcuOw lgeou p+WymxKAJD8JdNjQEd2mOIGf9/8JRZ7Coxk9SBioWz6ks25nY9OTZ5Da27ExqxXA/ WxEg==
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=mime-version:message-id:sender-to:subject:from:date :domainkey-signature:dkim-signature; bh=E3jq0JUnzM+qgNN5xbDwv7zDnBLJ9Qw LNK0BmkJUL; h=dmfcYm5f8yB85o7PJJZgD684tXo9Htp64Jazk4btw=; b=AZJLMJDY2u44h2AKkUEST7MvZyvp1PnImcAwQ3Uscv682KLLwhqwmDwG0FR H2VPGRhUJK3SfNSmCXaIFQY82aECoG HsGoWwXoIbJQs8qgDYEhXTC2LZYaaY zDqyKJ/JhVZPDSSYbJEgTXDHhT9H4Fw5UWOCeEIoP/ooNpH3aKQm1rCsdENQJDM ZnY9KR+DeO04Zno1nZhhQmMDGbuJu7p/g9CDI+qzJ9bkqMMY2qtGC0NjhW dBr7MYOh /guHpS/ZagMnT9W4TbOJGBqRwuc+sGIXanTztjgntQdkpallKYNlaX3hivGAItbl xqXg=; dara=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@wikezjska.freetutorial.eu.org headers.s=selector5 header.b=EJObzZio; spf=pass (google.com: domain of o2f25y8j3ysz@freetutorial.eu.org designates 109.228.46.58 as p ermitted sender) smtp.mailfrom=o2f25y8j3ysz@freetutorial.eu.org
Return-Path	<o2f25y8j3ysz@freetutorial.eu.org>
Received-SPF	pass (google.com: domain of o2f25y8j3ysz@freetutorial.eu.org designates 109.228.46.58 as permitted sender) client-ip=109.228.46.58;
Authentication-Results	mx.google.com; dkim=pass header.i=@wikezjska.freetutorial.eu.org headers.s=selector5 header.b=EJObzZio; spf=pass (google.com: domain of o2f25y8j3ysz@freetutorial.eu.org designates 109.228.46.58 as permi tted sender) smtp.mailfrom=o2f25y8j3ysz@freetutorial.eu.org
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; s=selector5; d=wikezjska.freetutorial.eu.org; h=Date:From:Subject:To:Sender:Message-ID:MIME-Version:Content-Type; i=mewofu4k3@wikezjska.freetutorial.eu.org; bh=E3jq0JUnzM+qgNN5xbDwv7zDnBLJ9QwL NK0BmkJUL; b=EJObzZioGCaVKF58XvzDTzMGv4LHuGKw8H7IS6QJ3LgqXf+NqJ+bfQkw5313wPX8KUTGIPAZkMP u+6xZTWw7wTqLab0J5mYNXaDuSht149/ICxgG E7Ad/zadfdOn72Cje8mRCUlgZoxh9vAHTMe KJ/vVv1Mk8NscSNwqTw=
DomainKey-Signature	a=rsa-sha1; c=nores; q=dns; s=m1; d=wikezjska.freetutorial.eu.org; b=fzaMlUXK4SogbzRUJ85d5EXS8NkQaKnuDx7/CjKwU8oeaCkoZKxIZhyADSfC7thwZr1pf briffgypP/xuH3Ly0SMOnKntZr3ajSht7xzeYPL VDMYKakVsoyjpU1gV9e2uTaK9LChLW6tims oEBG77VZDYAuhRD0eB1hKJZxyEmJG36quQJlVvVdOJa5Ge5Ta5j6EK/Kas/8TokJ82MuKok Wty6FyaXryL9H9byqrQnBjhYCBbeCofmSfoXIMB0nK7t0ot7VW441 L5XAgAmCBLbmf6u8 eb9ryvM80h2eDpKmneyUEQAJPhq1/MTnJQ=;
Date	Wed, 29 Jul 2026 18:38:43 +0000 (UTC)
From	Vivinhomeseurity <nooreply@1109532955108.zvxxqqcbsilvtog.uk>
Subject	Re: .Last_Reminder_Today! Wed, Jul 29 2026.
To	
Sender	rdicnqe <9mewofu4k3@wikezjska.freetutorial.eu.org>
Message-ID	<xlf0qgin.rdicnqe.rdicnqe.javamail.tomcat@pdr8-services-05v.prod.rdicnqe.org>
MIME-Version	1.0
Content-Type	multipart/report; boundary="000000000000SGCNuB7Qm6umj710eBSQB"; report-type=delivery-status

5.3 Email Authentication Analysis

Email authentication mechanisms were examined to determine whether the message had been legitimately sent by the claimed organization.

Results:

Authentication	Result
SPF	FAIL
DKIM	FAIL
DMARC	NONE

Figure 3. SPF, DKIM, and DMARC authentication results

When analyzing the review of the response, I noticed that the SPF alignment was categorized as “domain not found in SPF”, meaning there exists no record of this when it was looked up

As I investigated why the DKIM failed, it mentioned “DKIM Syntax Invalid” suggesting that it cannot be validated

Lastly, for DMARC, it stated that there was “No record found”. The sender domain did not publish a DMARC policy that could be evaluated. This reduces the ability of receiving mail systems to authenticate messages claiming to originate from the domain.

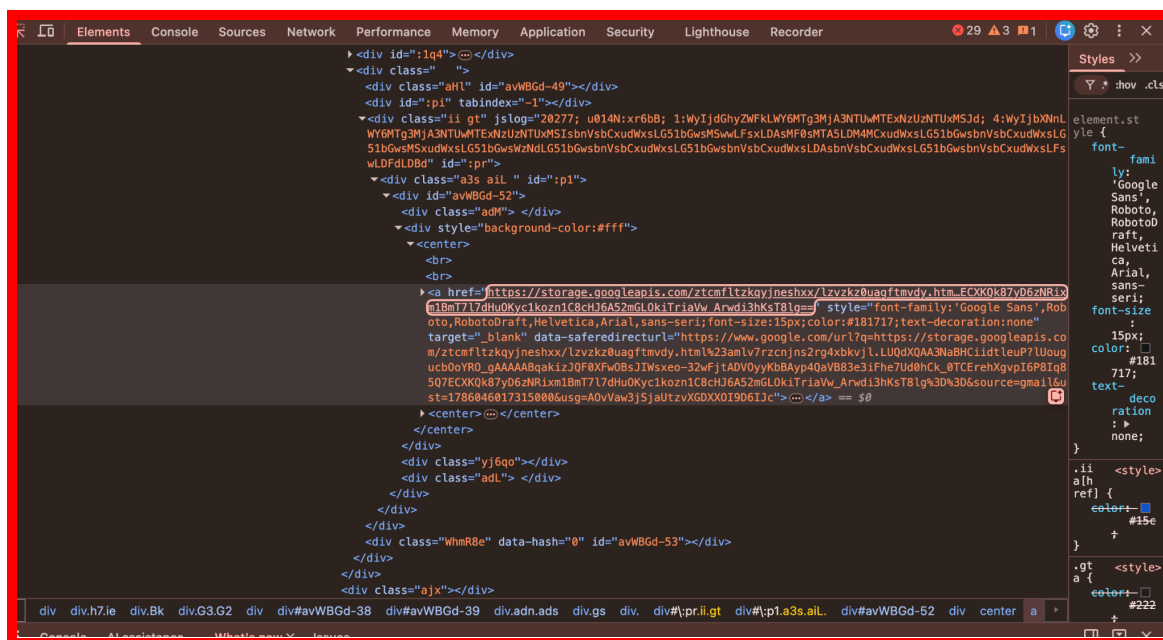
5.4 Embedded Link Analysis

The embedded link in the offer button was not opened for investigation

As stated, the email instructed the user to click on the button to claim the offer

Since clicking on this already suspicious email was not ideal, the embedded hyperlink showed a URL that linked directly to “googleapis”. We can assume that they used it as a means to bypass security filters and trick users.

Figure 4. Embedded hyperlink analysis identified a suspicious URL



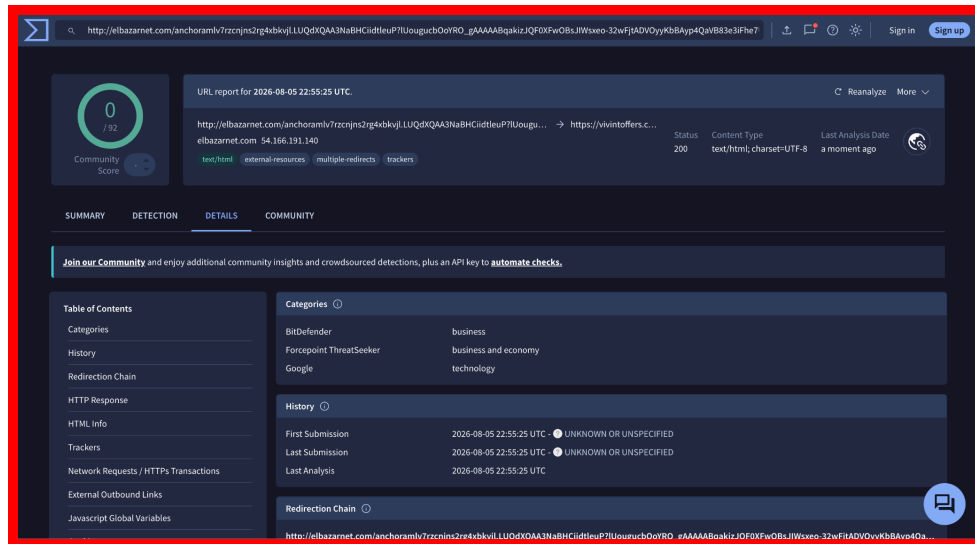
5.5 Threat Intelligence Investigation

The sender domain and sending IP address were sent to VirusTotal.

Threat intelligent searches were performed to determine whether the domain or IP address had been associated with malicious activity

Although the result of the scan came out as 0/92, meaning there was no detection of any virus. However, that doesn't completely mean that it is 100% clear, and based on the other analysis done in this project, that is certainly not the case.

Figure 5. Threat Intelligence investigation using



5.6 Indicators of Compromise (IOCs)

IOC Type	Value	Analysis
Sender Email	nooreply@4695370485734.zbsbwglwokmebkgigh.org.uk	Incorrect domain from the company

SPF	FAIL	Meaning there exists no record of this when it was looked up
DKIM	FAIL	Suggests that it cannot be validated
DKIM	FAIL	It cannot be validated
DMARC	NONE	Reduces the ability of receiving mail systems to authenticate messages claiming to originate from the domain.
Embedded URL	An image taking a user to an external site	Linked to google.apis not to a site that is associated with the company

The combination of these indicators strongly suggests that the message was designed to steal user credentials.

5.7 Risk Assessment

Based on the collected evidence, the investigated email presents a **High Risk**.

Evidence Supporting the Risk Rating

- Unknown email domain.
- Failed SPF authentication.
- Failed DKIM signature.
- Missing DMARC validation.
- Suspicious URL.

The investigation determined that users should not interact with the email.

Security Recommendations

The following recommendations are to be followed..

- Email Header Analysis
- Phishing Investigation
- Open-Source Intelligence (OSINT)
- Threat Intelligence
- SPF Analysis
- DKIM Validation
- DMARC Evaluation
- IOC Identification
- Security Reporting
- Risk Assessment
- Incident Investigation